

Networking Basics for Cyber Security

Introduction

Basic Network Flow Diagram

Networking forms the backbone of modern computing systems. Every application, website, email, and online service relies on networks to transmit data. In cybersecurity, understanding networking basics is extremely important because most cyber attacks either target network communication or use networks as a medium to spread.

Networking Basics for Cyber Security focuses on how data travels across networks, how devices communicate, and how attackers exploit network weaknesses. By analyzing network traffic, security professionals can detect malicious activities, unauthorized access, data leakage, and intrusion attempts.

This document explains fundamental networking concepts from a cybersecurity perspective using packet analysis tools such as Wireshark. The content is written in a beginner-friendly and structured format suitable for internship and academic evaluation.*

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Why Networking Knowledge Is Important in Cyber Security

- Most cyber attacks occur over networks.
- Hackers exploit unsecured network communication.
- Understanding networks helps detect suspicious traffic.
- Network monitoring helps prevent data breaches.
- Firewalls, IDS, and IPS work based on networking principles.
- Security analysts rely on packet analysis to investigate incidents.

Real-Life Example:

If sensitive data is transmitted without encryption, an attacker on the same network can capture it using packet sniffing tools.

Tools Used

Tools Used

Primary Tool

- **Wireshark**
- Popular network packet analyzer
- Captures and inspects live network traffic
- Widely used by cybersecurity professionals

Alternative Tools

- **tcpdump**
- Command-line based packet capturing tool
- Commonly used on Linux servers
- **Microsoft Network Monitor**
- Windows-based network analysis tool
- Used for troubleshooting and traffic analysis

These tools help analyze how data flows across networks and identify security issues.

Basic Networking Concepts

Basic Networking Concepts

What Is a Network?

- A network is a group of connected devices.
- Devices communicate using wired or wireless connections.
- Networks allow data sharing and resource access.
- Examples include LAN, WAN, and the Internet.

Real-Life Example:

A home Wi-Fi network connects mobile phones, laptops, and smart TVs.

IP Address

- An IP address uniquely identifies a device on a network.
- IPv4 uses 32-bit addresses (e.g., 192.168.1.1).
- IPv6 uses 128-bit addresses for better scalability.
- IP addresses help route data correctly.

Cyber Security Importance:

Attackers are often identified using suspicious IP addresses.

MAC Address

- A MAC address is a hardware address.
- It uniquely identifies a network interface.
- Used within local networks.
- Hardcoded into network devices.

Example:

MAC filtering is used to allow only trusted devices.

Networking Protocols Important for Cyber Security

TCP Three Way Handshake Diagram

This diagram illustrates the TCP three-way handshake (SYN, SYN-ACK, ACK) used to establish reliable connections, and contrasts it with UDP communication which does not use connection establishment.*

Networking Protocols Important for Cyber Security

TCP (Transmission Control Protocol)

- Connection-oriented protocol
- Ensures reliable data transmission
- Used for web browsing, email, file transfer
- Performs error checking and retransmission

Security Note:

Attackers exploit TCP using SYN flood attacks.

UDP (User Datagram Protocol)

- Connectionless protocol
- Faster but less reliable
- Used for video streaming and DNS
- No error correction

Security Risk:

UDP is commonly used in DDoS attacks.

HTTP and HTTPS

- HTTP transmits data in plain text.
- HTTPS encrypts data using SSL/TLS.
- HTTPS prevents data interception.

Real-Life Example:

Logging into a website without HTTPS can expose passwords.

What Is Packet Sniffing?

This diagram illustrates how network packets are captured from live traffic using a packet sniffing tool such as Wireshark.

What Is Packet Sniffing?

Packet sniffing is the process of capturing and analyzing network packets.

- Used by security professionals for monitoring
- Used by attackers for spying
- Helps detect malicious traffic
- Reveals source, destination, and payload

Cyber Security Use Case:

Identifying suspicious login attempts or data leaks.

Wireshark Tool Overview

Wireshark User Interface

This image represents the Wireshark window showing captured packets, protocol details, and packet bytes.*

Wireshark Tool Overview

Wireshark is a graphical network packet analyzer.

Features of Wireshark

- Captures live network traffic
- Supports multiple protocols
- Displays packet details clearly
- Provides filtering options

Why Wireshark Is Important

- Helps understand network behavior
 - Detects anomalies
 - Used in incident response
 - Essential for network forensics
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Capturing Packets Using Wireshark

Capturing Packets Using Wireshark

Steps to Capture Packets

- Open Wireshark
- Select a network interface
- Click Start Capture
- Generate network traffic
- Stop capture and analyze packets

Example:

Opening a website generates HTTP/HTTPS packets.

Understanding Wireshark Packet Details

Understanding Wireshark Packet Details

Packet Information Includes

- Source IP address
- Destination IP address
- Protocol used
- Packet length
- Payload data

Packet Layers

- Frame layer
- Network layer
- Transport layer
- Application layer

Understanding layers helps locate security issues.

Filters in Wireshark

Filters in Wireshark

Filters help focus on specific traffic.

Common Filters

```
ip.addr == 192.168.1.1
```

```
tcp
```

```
http
```

Explanation: - Filters reduce noise - Makes analysis faster - Helps isolate malicious traffic

Networking Attacks Detectable Using Packet Analysis

This diagram shows how an attacker intercepts communication between a client and server without their knowledge.

Networking Attacks Detectable Using Packet Analysis

- Man-in-the-Middle attacks
- Packet sniffing
- DNS spoofing
- ARP poisoning
- Unauthorized access attempts

Example:

Repeated login failures from the same IP indicate brute-force attacks.

Cyber Security Best Practices for Network Security

- Use encrypted protocols (HTTPS, SSH)
 - Monitor network traffic regularly
 - Use firewalls and IDS
 - Avoid open and unsecured Wi-Fi
 - Update network devices
 - Restrict unused ports and services
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Simple Real-World Security Scenario

Simple Real-World Security Scenario

A user connects to public Wi-Fi and logs into a website using HTTP. An attacker on the same network captures packets and steals login credentials using a packet sniffer.

If Network Security Is Applied: - HTTPS encrypts traffic - Packet data remains unreadable - User credentials stay protected

Practical Networking Tasks & Observations

This section documents the practical steps performed using Wireshark along with simple observations, as required for the internship task.

1. Learning Basic Networking Concepts (IP, MAC, DNS, TCP/UDP)

- IP Address identifies a device on a network uniquely.
- MAC Address is a hardware-level identifier assigned to network interfaces.
- DNS translates domain names into IP addresses.
- TCP provides reliable, connection-oriented communication.

- UDP provides fast, connectionless communication.
- These concepts form the foundation for understanding network traffic analysis.

Observation:

All captured packets in Wireshark contain source and destination IP and MAC information.

2. Installing Wireshark and Capturing Live Network Traffic

- Wireshark was installed on the system successfully.
- Network interface (Wi-Fi/Ethernet) was selected.
- Live packet capture was started.
- Normal browsing activity generated network packets.
- Packets were captured in real time.

Observation:

Large numbers of packets are generated even during simple internet usage.

3. Filtering Packets by Protocol (HTTP, DNS, TCP)

```
http
dns
tcp
```

- Filters were applied to isolate specific traffic.
- HTTP filter showed web traffic.
- DNS filter showed domain name resolution.
- TCP filter displayed reliable connection traffic.
- Filtering reduced unnecessary packet noise.

Observation:

Filters make packet analysis easier and faster.

4. Observing the Three-Way TCP Handshake

- TCP connection starts with SYN packet.
- Server responds with SYN-ACK.
- Client replies with ACK.
- This establishes a secure TCP session.
- Visible clearly in Wireshark capture.

Observation:

Every TCP connection follows the same handshake pattern.

5. Identifying Plain-Text Traffic vs Encrypted Traffic

- HTTP traffic is visible in plain text.
- HTTPS traffic appears encrypted.
- Packet payloads differ clearly.
- Encrypted traffic hides sensitive data.
- Plain-text traffic exposes information.

Observation:

HTTPS is much safer than HTTP for data transmission.

6. Capturing and Analyzing DNS Queries

This diagram shows how a DNS query is sent from a client to a DNS server and how the corresponding IP address is returned before a website is loaded.

- DNS packets were filtered using `dns`.
- Queries show requested domain names.
- Responses show resolved IP addresses.
- DNS uses UDP by default.
- DNS traffic occurs before website loading.

Observation:

Every website visit triggers multiple DNS queries.

7. Saving Packet Captures for Analysis

- Captured packets were saved using `.pcap` format.
- Saved files can be reopened later.
- Useful for forensic analysis.
- Helps in documentation and reporting.
- Enables offline packet study.

Observation:

Saved captures are helpful for future reference and audits.

8. Writing Observations in Simple Language

- Observations were written clearly.
 - Technical jargon was minimized.
 - Each task outcome was documented.
 - Focus was on understanding rather than complexity.
 - Suitable for academic evaluation.
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Final Outcome / Learning Summary

By completing these practical networking tasks, a clear understanding of basic networking concepts and packet analysis was achieved. Wireshark helped visualize real network communication, identify encrypted and plain-text traffic, analyze DNS queries, and understand TCP connections. This practical knowledge strengthens the foundation required for cybersecurity analysis, network monitoring, and incident investigation.